YONATHAN WONDOSEN

Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 5+ years of production experience designing, building, and scaling secure, high-performance web applications and enterprise platforms. Proficient in React, Next.js, TypeScript, Python (FastAPI, Flask), C#/.NET, and Laravel, with expertise in AI-powered applications, RAG pipelines, vector search, microservices optimization, and financial system integrations. Experienced in translating complex business requirements into reliable, maintainable, and production-ready software solutions.

WORK EXPERIENCE

Fullstack Software Engineer | VELKAP Ltd

London, UK (Remote) | 2026 – Present (Contractual)

- Architecting and engineering an enterprise investment management platform for a UK-based fintech firm utilizing Laravel and Next.js.
- Designing high-throughput backend APIs for financial transaction processing and investment portfolio tracking.
- Developing highly responsive, accessible user interfaces using modern React patterns, TypeScript, and state management strategies.
- Ensuring strict compliance with UK financial regulatory guidelines and data security standards across all services.
- **Key Achievement:** Engineered a complex co-investor feature enabling multi-party equity distribution. Modeled relational database schemas and algorithmic allocation logic to automate participant share calculations, audit logging, and precise reporting.

AI Focus Back-end Engineer, RAG Development | Akilo Technology

Hong Kong, China (Remote) | 2025 – 2025 (Contractual)

- Architected a RAG AI chatbot integrating Large Language Models (LLMs) with MongoDB vector search via HNSW indexing.
- Engineered automated pipelines for embedding storage and semantic retrieval, supplying contextual grounding to LLM prompts.
- **Key Achievement:** Resolved retrieval accuracy bottlenecks by tuning HNSW parameters, optimizing vector embedding strategies, and introducing semantic document chunking—substantially decreasing LLM hallucination and improving contextual relevance.

Fullstack Software Engineer | EtPay-Rent PLC

Addis Ababa, Ethiopia | March 2025 – Nov 2025 (Full-time / Hybrid)

- Led full-stack engineering for a microservices-based real estate and rent collection application, ensuring high availability and system reliability.
- Integrated local and international digital payment gateways, including Chapa/ Telebirr, establishing fault-tolerant transaction workflows.
- Investigated and resolved core payment integration edge cases, directly enhancing payment processing success rates.
- Evaluated third-party integrations and modern architectural paradigms to drive continuous platform optimization.
- **Key Achievement:** Boosted application throughput and reduced response latency by refactoring resource-intensive backend modules and optimizing database query executions.

Technology Officer | Ethiopian Kidney Care NGO

*Addis Ababa, Ethiopia | 2024 – Present (**Volunteer**)*

- Directed technical initiatives and managed core cloud services, and digital platforms.
- Designed and deployed an automated health screening web application, streamlining patient data validation and campaign analytics.
- Expanded web application capabilities by building user-focused features and responsive user interfaces.
- **Key Achievement:** Centralized fragmented software systems by establishing unified engineering standards, automated deployment processes, and structured software documentation.

FEATURED PROJECTS

Investment Management Platform (velkap.com)

Fintech investment platform facilitating automated portfolio tracking, financial compliance, and multi-party equity distribution through custom co-investor allocation logic and auditing.

- **Tech Stack:** C#/.NET, Laravel, Next.js, TypeScript, PostgreSQL, REST API

AI-Powered Electronic Medical Record System (bego.et)

Comprehensive electronic medical records platform engineered for healthcare providers to streamline clinical workflows, branch management, and patient care analytics.

- Tech Stack: React, Flask, REST API, PostgreSQL

Retrieval Augmented Generation (RAG) AI Chatbot (akilo technology)

High-performance retrieval-augmented generation engine pairing LLMs with MongoDB vector search for contextual response synthesis.

- **Tech Stack:** Python, Flask, OpenAI API, MongoDB, HNSW, Vector Search

Property Management System (EtPay Rent)

End-to-end real estate solution featuring automated rental payment processing, tenant management, and maintenance dispatch tracking.

- Tech Stack: React, Java Spring Boot, PostgreSQL, REST API

Better Auth – OAuth Implementation

Robust authentication architecture delivering protected OAuth login workflows, automatic session persistence with Better Auth, and synchronized Prisma schema integration.

- Tech Stack: Node.js, Next.js, Prisma, OAuth, Better Auth

Payment Gateway Integration (Chapa / Telebirr / Stripe)

Enterprise payment gateway module providing robust transaction reconciliation, webhooks, and secure API handling for payment providers.

- **Tech Stack:** CHAPA_API, REST API, Laravel, MySQL

AI-Driven MicroFinance Solution

Fintech risk assessment engine implementing AI-driven credit scoring models based on verified identity metrics and behavioral datasets.

- Tech Stack: Python, React, AI, REST API, JWT, Chapa Payment Gateway Integration

SKILLS & COMPETENCIES

Domain	Core Skills & Technologies
Architecture & DevOps	Enterprise System Design, Microservices, RAG Development, Vector Search (HNSW, MongoDB Vector Search), Docker, Kubernetes, PM2, CI/CD Pipelines, VPS Infrastructure, Payment Integration
Frontend	React, Next.js, TypeScript
Backend	Python (FastAPI, Flask), C#/.NET , Laravel, Node.js Express, Java Spring Boot, REST APIs, GraphQL
Databases	PostgreSQL, MySQL, MongoDB, Prisma ORM
Professional	Project Management, Team Leadership, Time Management, Effective Communication

EDUCATION

BSc in Computer Science

Unity University – Addis Ababa, Ethiopia (2022 – 2026)

LANGUAGES

- **English:** Professional Working Proficiency
- **Amharic:** Native
- **Oromo:** Basic Proficiency